

Deepak Kumar Somasundaram

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EDUCATION

Master of Science in Information Systems

Northeastern University, USA

Sep 2024 - Apr 2026

Focus Areas: Full-stack Development, Cloud & DevOps, AI Engineering

TECHNICAL SKILLS

Programming: Java, Python, C#, JavaScript, TypeScript, Bash

Database Systems: MySQL, PostgreSQL, MongoDB, DynamoDB, Redis

Web Development: HTML, CSS, React, Angular, REST APIs, .NET Core, Spring Boot, FastAPI, Node.js

Cloud & DevOps: AWS, Azure, Azure DevOps, Git, GitHub, GitHub Actions, Docker, Terraform

AI Development: LLMs, RAG, LangGraph, OpenAI API, Anthropic API, AWS Bedrock, MCP

PROFESSIONAL EXPERIENCE

Software Engineer, Prodex Technologies

Feb 2023 - Jul 2024

Skills – Java, Spring Boot, Hibernate, JavaScript, React, MySQL, Docker, CI/CD

- Architected **5+** features per quarter using **Java** and **Spring Boot** within a **microservices** backend, enabling rapid feature development and service integration.
- Modernized **15+** legacy **GWT** screens by migrating modules to **React** and **TypeScript**, partnering with the Customer Success team to address **client feedback** and deliver a more responsive, maintainable frontend.
- Reduced backend regression issues by **40%** before sandbox deployment through comprehensive unit and integration testing with **JUnit**.
- Automated **Docker image** deployments across AWS **ECR/ECS** via **CI/CD** workflows, ensuring **2+** reliable releases per week.
- Documented **10+ new APIs** using **AI-generated API documentation**, improving issue traceability, future maintenance, and developer onboarding.

Software Engineer, HealthOneCloud

Jan 2022 - Jan 2023

Skills – Microsoft Azure, .NET Core 8, C#, Angular, SQL Server

- Developed backend APIs in **.NET Core 8** and **Entity Framework** for a healthcare SaaS platform, supporting **10+** distinct medical billing workflows internally routed through **Azure API Management**.
- Optimized **T-SQL stored procedures** in **MS SQL Server** for billing data retrieval, improving query performance by **25%** while preserving data integrity across releases.
- Migrated a client-requested feature across **Angular** modules to a new screen, preserving existing functionality while simplifying the user navigation flow.
- Led knowledge transfer sessions for **automation solutions**, enabling team members to understand the implementation, business context, and maintenance workflow.
- Orchestrated a **Node.js** log-parsing solution with **Azure Logic Apps** after collaborating with the **QA** team to analyze manual log review workflows, automating Microsoft Teams notifications and reducing coordination effort by **75%**.

PROJECTS

TerraGuard — Multi-Agent IaC Governance System

Apr 2026

- Built a multi-agent IaC governance system using **LangGraph** and **Agentic AI** where reviewer and fixer agents autonomously detect policy violations and generate corrected code, cutting manual PR review effort by **50%**.
- Engineered hybrid pipeline with deterministic checks and **LLM-driven** code generation (**OpenAI**, **RAG**, **ChromaDB**), achieving **~90%** first-pass fix accuracy for **IaC** policy violations.
- Integrated GitHub and Gmail **MCP servers**, enabling autonomous agent actions like opening fix PRs and sending notifications, reducing violation to resolution time from hours to **under 2 minutes**.

LLM-Assisted Pipeline Debugger

Jan 2026

- Designed serverless Python bot using **AWS Lambda** and **API Gateway** to intercept **CI/CD** failures via GitHub webhooks, reducing manual debugging time by **2 hours** per day.
- Constructed a prompt pipeline that cleans raw logs and pairs code diffs with structured context for **AWS Bedrock** (Claude) analysis, delivering root cause analysis with fixes in under **10 seconds**.
- Containerized the application using **Docker** and synchronized **AWS ECS** deployments through **GitHub Actions CI/CD**, enabling reliable production releases.

Cloud-Native Full Stack Platform on AWS

Oct 2025

- Spearheaded the development of a **JWT**-authenticated PR monitoring platform using **Python (FastAPI)** and **React**, implementing **OTP** authentication with **AWS SES** for **secure** user access.
- Optimized **Redis caching** with TTL-based expiry, reducing API response time from **1470ms to 20ms (98.6% faster)** and implemented rate-limited OTP storage for secure authentication flows.
- Architected cloud infrastructure on **AWS** using **Terraform**, provisioning **VPC**, **ECS Fargate**, **ALB**, **RDS**, **ElastiCache**, **S3 + CloudFront**, **Route 53**, **KMS**, with **CloudWatch** monitoring and centralized logging.
- Automated end-to-end application delivery using **GitHub Actions CI/CD**, orchestrating zero-downtime deployments of frontend and backend services across AWS infrastructure.